LAB # 09: Depth Measurement Using Stereo Images

Lab Objective:

The objective of this lab is to understand and implement depth measurement using stereo images from calibrated cameras. Students will learn how to compute disparity maps and convert them into depth maps using intrinsic and extrinsic camera parameters obtained from calibration.

Lab Description:

1. Stereo Vision Setup

A stereo vision system consists of two cameras placed at a fixed distance from each other, known as the baseline (B). These cameras capture images of the same scene from slightly different viewpoints, similar to how human eyes perceive depth.

Due to this difference in viewpoint, the same object appears at slightly different horizontal positions in the two images. This positional difference is the key to estimating depth. Intuitively, objects that are closer to the cameras appear with a larger shift, while distant objects show very little shift.

To ensure reliable matching between the two images, stereo systems rely on epipolar geometry, which defines the geometric relationship between the two camera views. After applying a process called rectification, corresponding points in both images lie on the same horizontal lines. This simplifies the correspondence problem by reducing the search for matching points from a 2D area to a 1D horizontal line.

2. Disparity and Depth Relationship

Disparity (d) is defined as the horizontal shift of a point between the left and right images. For each pixel in the left image, we search for the corresponding pixel in the right image and measure how far it has moved horizontally.

This disparity is directly related to the depth of the object. The relationship is given by:

$$Z = \frac{f \cdot B}{d}$$

where:

- Z is the depth (distance from the camera),
- f is the focal length (in pixels),
- B is the baseline distance between the cameras,

- d is the disparity.

This equation shows that depth is inversely proportional to disparity. In simple terms:

- Large disparity shows object is close
- Small disparity shows object is far

This relationship allows us to convert pixel-level differences into real-world distance measurements.

3. Steps in Stereo Depth Estimation

Stereo depth estimation is a pipeline, where each step depends on the previous one. Missing or poorly performing any step will affect the final depth accuracy.

a) Camera Calibration

Camera calibration is the foundation of the entire system. It involves estimating:

- Intrinsic parameters: focal length, principal point, lens distortion
- Extrinsic parameters: rotation and translation between the two cameras

Real-world cameras introduce distortions and are rarely perfectly aligned. Calibration corrects these issues and ensures that both images follow a consistent geometric model. Without proper calibration, depth estimation becomes unreliable.

b) Image Rectification

Rectification transforms the stereo images so that corresponding points lie on the same horizontal lines. This step removes vertical misalignment caused by camera placement.

The main benefit of rectification is computational efficiency. Instead of searching for matches across the entire image, we only need to search along horizontal lines. This significantly reduces complexity and improves matching accuracy.

c) Disparity Map Computation

In this step, we compute the disparity map, which is a representation of pixel shifts across the entire image.

For each pixel in the left image, we search along the corresponding horizontal line in the right image to find the best match. The difference in their horizontal positions is recorded as the disparity value.

This process is typically performed using:

- Block matching (comparing small pixel patches)
- Semi-global matching (more robust and accurate)

The output is a disparity map, where:

- Higher values (brighter pixels) indicate closer objects
- Lower values (darker pixels) indicate distant objects

This map serves as the intermediate step before computing actual depth.

Why Stereo Depth Measurement?

Stereo vision is widely used because it offers a balance between performance and cost:

- Passive Sensing: It does not require emitting signals like LiDAR; it relies only on captured images.
- Low Cost: Standard cameras are sufficient, making it accessible and scalable.
- Real-Time Capability: With optimized algorithms, depth can be estimated quickly for applications like robotics and autonomous systems.
- Flexibility: It can be deployed in various environments without specialized hardware.

Challenges

While stereo vision is powerful, it comes with practical limitations:

- Textureless Regions: Uniform surfaces lack distinctive features, making it difficult to find correct matches.
- Occlusions: Some parts of the scene may be visible in one image but not the other, leading to missing or incorrect disparity values.
- Lighting Variations: Differences in illumination between the two images can affect matching accuracy.
- Computational Cost: Dense disparity estimation for high-resolution images can be computationally expensive without optimization.

Understanding these challenges helps in interpreting results and improving system performance.

Applications

Stereo depth estimation is used in many real-world systems:

- Autonomous Driving: Estimating distances to vehicles, pedestrians, and obstacles.
- 3D Reconstruction: Generating 3D models from multiple images.
- Augmented Reality: Accurately placing virtual objects in real-world scenes.
- Robotics: Enabling navigation and obstacle avoidance.

Task:

Design and implement a stereo vision pipeline to estimate depth from a pair of images. The system should detect and match keypoints between stereo images, estimate the geometric relationship between the cameras, recover camera pose, reconstruct 3D points using triangulation, and generate a depth map for visualization.